

PROBLEM FORMULATION

Unbalanced Schrödinger Bridge Problem

$$\min_{u_t, g_t} \int_0^1 \mathbb{E}_{p_t} \left[\frac{1}{2} \|u_t(X_t)\|_2^2 + V_t(X_t) \right] w_t(X_t) dt \quad \text{s.t.} \quad \begin{cases} dX_t = u_t(X_t)dt + \sigma dB_t \\ X_0 \sim \pi_0, \quad X_1 \sim \pi_1 \\ w_0 = w_0^*, \quad w_1 = w_1^* \end{cases}$$

Branched Conditional Stochastic Optimal Control

Primary Branch

$$\min_{\{u_{t,k}, g_{t,k}\}_{k=0}^K} \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_{1,0}) \sim \pi_{0,1,0}} \int_0^1 \left\{ \mathbb{E}_{p_{t,0,1,0}} \left[\frac{1}{2} \|u_{t,0}(X_{t,0})\|_2^2 + V_t(X_{t,0}) \right] w_{t,0} \right\} dt \\ + \sum_{k=1}^K \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_{1,k}) \sim \pi_{0,1,k}} \int_0^1 \left\{ \mathbb{E}_{p_{t,0,1,k}} \left[\frac{1}{2} \|u_{t,k}(X_{t,k})\|_2^2 + V_t(X_{t,k}) \right] w_{t,k} \right\} dt$$

Secondary Branches

Subject to: $w_{t,0} = 1 + \int_0^t g_{s,0}(X_{s,0})ds \quad w_{t,k} = \int_0^t g_{s,k}(X_{s,k})ds$

$$dX_{t,k} = u_{t,k}(X_{t,k})dt + \sigma dB_t, \quad X_0 = \mathbf{x}_0, \quad X_{1,k} = \mathbf{x}_{1,k}, \quad w_{0,k} = \delta_{k=0}, \quad w_{1,k} = w_{1,k}^*$$

MULTI-STAGE TRAINING ALGORITHM

Stage 1: Learning Energy-Minimizing Branched Neural Interpolants

Train a **neural path interpolant** that learns the non-linear cost manifold, given a pair of endpoints.

$$\mathbf{x}_{t,\eta,k} = (1-t)\mathbf{x}_0 + t\mathbf{x}_{1,k} + t(1-t)\varphi_{t,\eta}(\mathbf{x}_0, \mathbf{x}_{1,k})$$

differentiate

$$\dot{\mathbf{x}}_{t,\eta,k} = \mathbf{x}_{1,k} - \mathbf{x}_0 + t(1-t)\dot{\varphi}_{t,\eta}(\mathbf{x}_0, \mathbf{x}_{1,k}) + (1-2t)\varphi_{t,\eta}(\mathbf{x}_0, \mathbf{x}_{1,k})$$

Stage 2: Learning Branched Neural Velocity Fields

We parameterize a set of neural drift fields that generate the mixture of bridges defined in Stage 1 by minimizing the **conditional flow matching loss**:

$$\mathcal{L}_{\text{flow}}(\theta) = \sum_{k=0}^K \int_0^1 \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_{1,k}) \sim \pi_{0,1,k}} \left\| \dot{\mathbf{x}}_{t,\eta,k}^* - \mathbf{u}_{t,k}^\theta(\mathbf{x}_{t,k}) \right\|_2^2 dt$$

Stage 3: Learning Growth Networks

We freeze the flow networks and train only the growth networks by minimizing the **weighted sum of the energy loss, weight matching loss and the total mass loss**.

$$\text{1) } \mathcal{L}_{\text{energy}}(\theta, \phi) = \int_0^1 \mathbb{E}_{\{p_{t,k}\}_{k=0}^K} \left\{ \left[\frac{1}{2} \|u_{t,0}^\theta(X_{t,0})\|_2^2 + V_t(X_{t,0}) \right] w_{t,0}^\phi + \sum_{k=1}^K \left[\frac{1}{2} \|u_{t,k}^\theta(X_{t,k})\|_2^2 + V_t(X_{t,k}) \right] w_{t,k}^\phi \right\} dt$$

$$\text{2) } \mathcal{L}_{\text{match}}(\phi) = \sum_{k=0}^K \mathbb{E}_{p_{1,k}} \left(w_{1,k}^\phi - w_{1,k}^* \right)^2, \quad \text{s.t.} \quad w_{1,k}^\phi = w_{0,k} + \int_0^1 g_{t,k}^\phi(X_{t,k})dt \quad \text{3) } \mathcal{L}_{\text{mass}}(\phi) = \int_0^1 \mathbb{E}_{\{p_{t,k}\}_{k=0}^K} \left[\left(\sum_{k=0}^K w_{t,k}^\phi - w_t^{\text{total}} \right)^2 + \sum_{k=0}^K \max(0, -w_{t,k}^\phi) \right] dt$$

Stage 4: Joint Training of Velocity and Growth Networks

In the final stage, we **unfreeze all parameters** and jointly train both the flow and growth networks using the same loss functions from Stage 3 together with a reconstruction loss.

BRANCHED SCHRÖDINGER BRIDGE MATCHING

How can we simulate energy-minimizing branching population dynamics by simulating only one sample from the initial distribution?

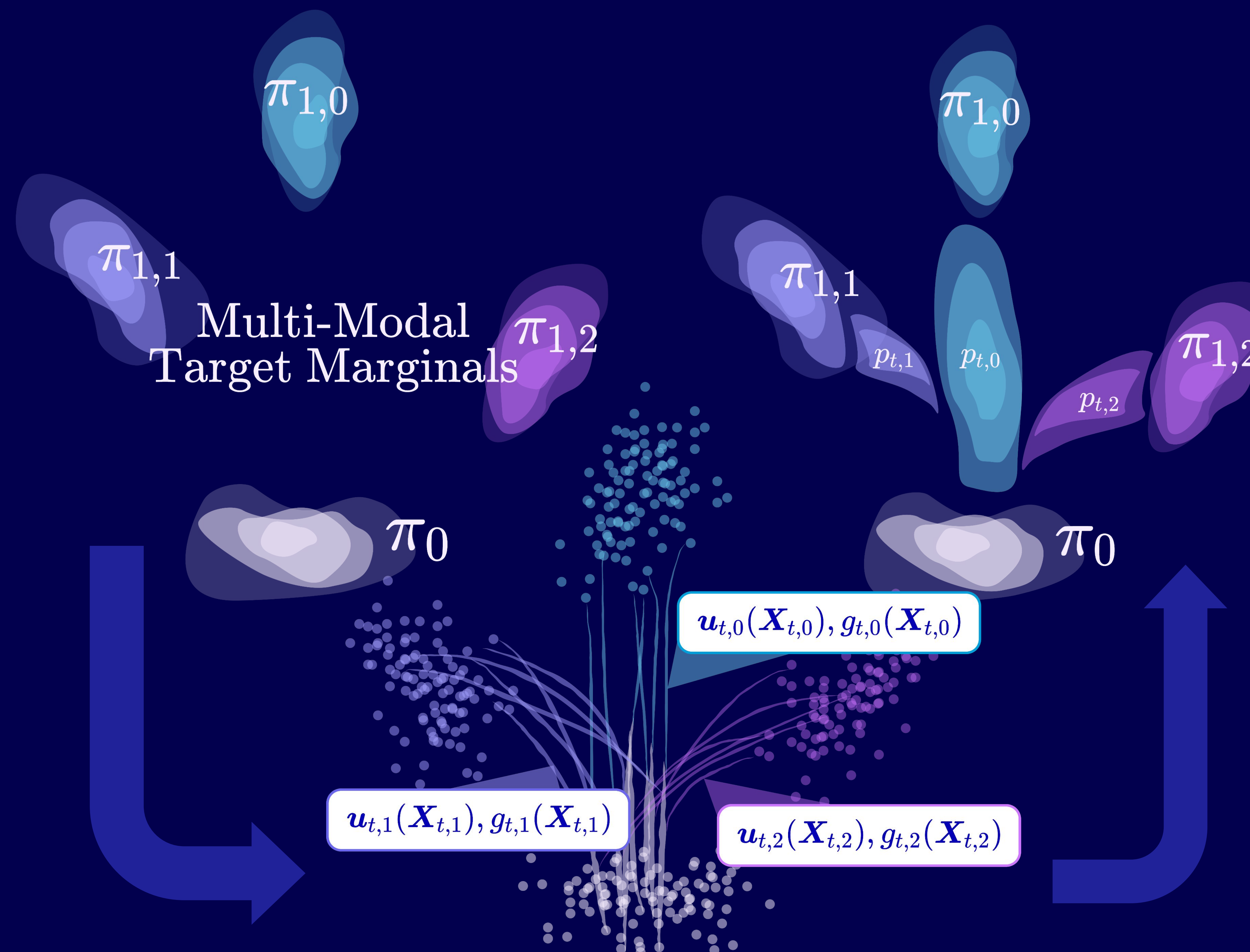
Simulation is often performed through many *independent* particle trajectories.

Reconstructing multi-modal distributions suffers from **mode collapse**.

Existing Schrödinger bridge and flow matching frameworks approximate multi-modal distributions by **simulating many independent particle trajectories**.

Since each particle is trained to traverse the low-energy paths to construct high-density terminal distributions, these methods are susceptible to **mode collapse**, with particles concentrating on dominant high-density modes or traversing only low-energy intermediate paths.

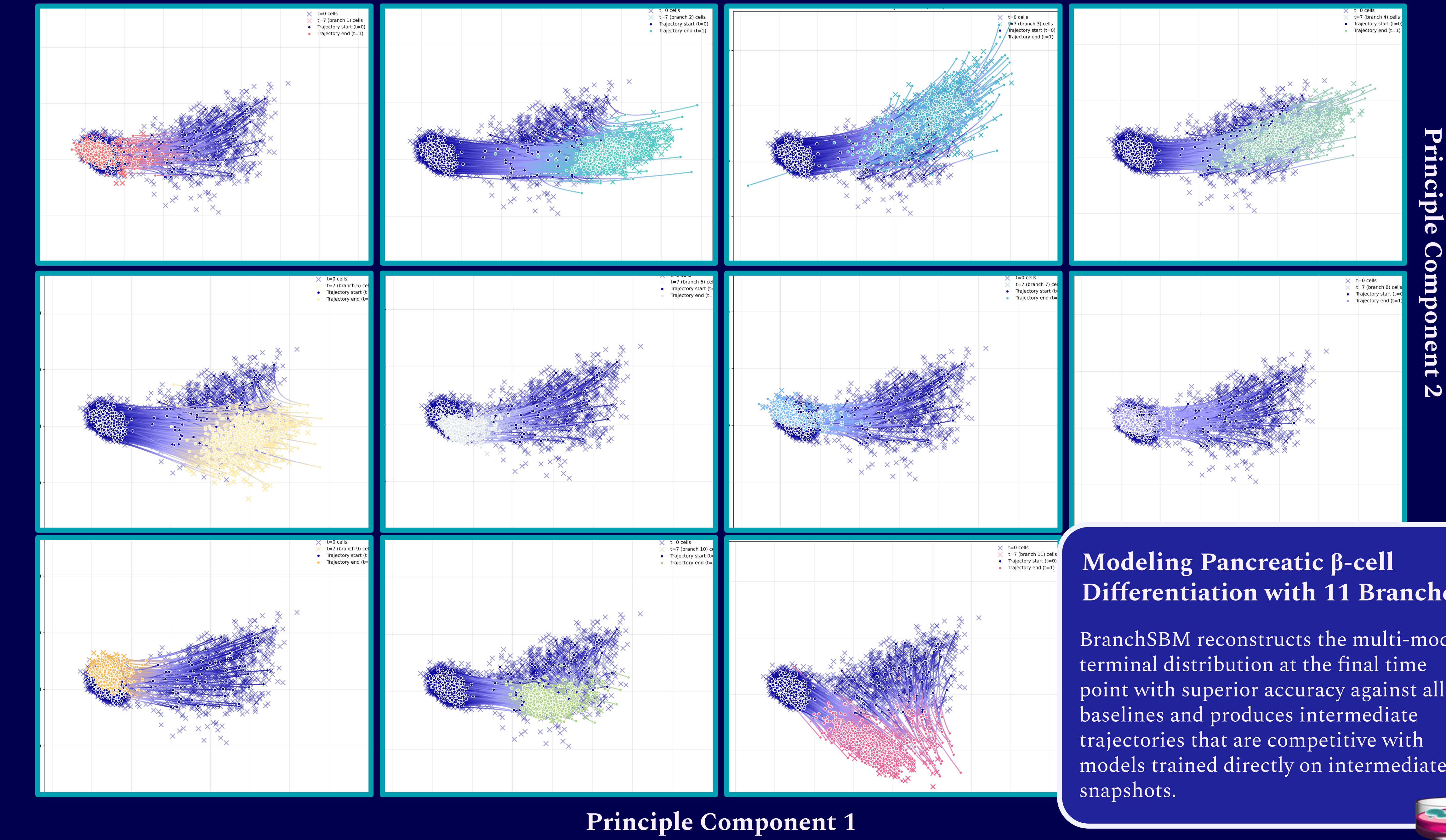
BranchSBM learns a set of *diverging velocity fields* to reconstruct multi-modal target distributions while simultaneously learning growth networks that allocate mass across branches.



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EXPERIMENTS

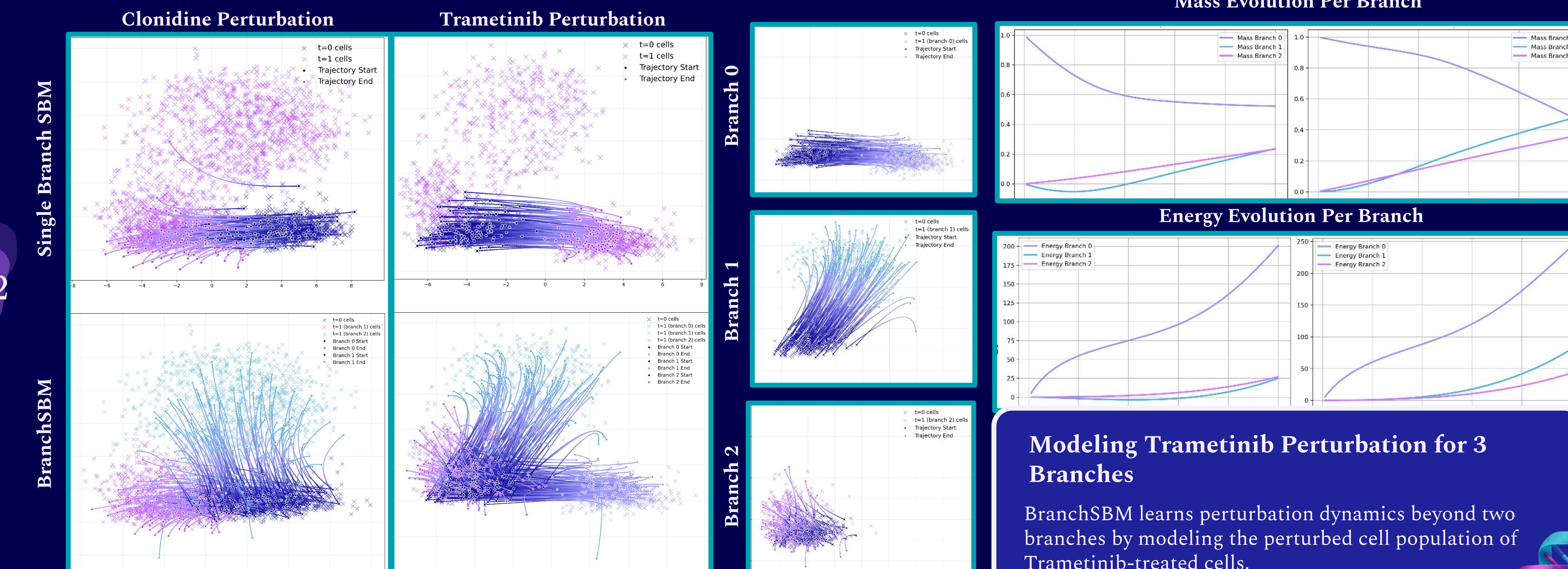
Modeling Single Cell Dynamics with BranchSBM



Modeling Pancreatic β -cell Differentiation with 11 Branches

BranchSBM reconstructs the multi-modal terminal distribution at the final time point with superior accuracy against all baselines and produces intermediate trajectories that are competitive with models trained directly on intermediate snapshots.

Simulating Cell Dynamics Under Drug Perturbation



Modeling Clonidine Perturbation Up to 150 PCs

BranchSBM reconstructs the diverging ground-truth distributions, whereas single-branch SBM fails to capture cells in higher dimensional principal components.

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Project Page



Paper



Code